

# Rahul Rustagi

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## Education

**Georgia Institute of Technology**, MS in Electrical and Computer Engineering Aug 2024 – May 2026  
Coursework: Multi-Robot Systems, Deep Learning for Robotics, PDEs in Image Processing GPA: 4.0/4.0

**Indian Institute of Technology (IIT) Kanpur**, BTech in Aerospace Engineering Aug 2020 – May 2024  
Coursework: Probabilistic Machine Learning, Reinforcement Learning, Optimal Control GPA: 9.18/10.0

## Work Experience

**End-to-End Collision Detection Models**, Symbolic – Boston, MA May 2025 – Aug 2025  
→ Built an ETL pipeline to collect 700 bot images from warehouses sites & used **Amazon Ion** logs to create depth ground truth  
→ Finetuned **KITTI** pre-trained Monocular Depth Estimation Models using ResNet, KeyPoint Detector and DINOv2 architecture  
→ Achieved **98.9%** Accuracy and **98.1%** recall of collision triggering events within 1 ft of proximity for the MonoDINO-DETR

**3D Scene Understanding & Object Tracking**, RAIL Lab – Georgia Tech Jan 2025 – May 2025  
→ Worked with **Amazon Lab126** to create an intelligent human-like perception algorithm for dynamic scene interpretability  
→ Implemented Khronos & Hydra on **Stretch RE2** from Hello Robot to create a **Spatio-Temporal** Scene Graph of environment

**Robot-Agnostic Advanced Navigation Assistance System (ADAS)**, IVALab – Georgia Tech Aug 2024 – Jan 2025  
→ Designed a hierarchal pipeline using multiple sensors like **IMU**, **LiDAR**, **RGB-Vision** for ensuring safe and reliable navigation  
→ Conducted sensor fusion using **GTSAM**, and evaluated **Hessian** matrix to analyze vision pose controllability in X,Y directions  
→ Devised a reliable hierarchal pipeline using **ORB\_SLAM2** as slow correction and **DSOL** as fast allowing for global navigation

**Vision-Based Autonomous Quadrotor Landing on Moving Ship**, Helicopter Lab – IIT Kanpur Aug 2023 – May 2024  
→ Published a tested pipeline to predict landing platform 6DoF motion, leveraging deep learning for UAV trajectory planning  
→ Used Fractal ArUco markers for estimating platform 6DoF pose at 60Hz using RGB stream from Intel Realsense D435i camera  
→ Deployed a **LSTM model** on **Jetson Nano**, integrating **TF-TRT & quantization** to effeciently predict future pose of platform

## Selected Publications | Scholar Profile

**Vision-Based Autonomous Ship Deck landing of Unmanned Aerial Vehicle using Fractal ArUco marker (2025)**  
C Prachand, *Rahul Rustagi*, R Shankar, J Singh, A Abhishek and K.S. Venkatesh | AIAA SciTech | doi: 10.2514/6.2025-2345

## Computer Vision Projects

**Deep Learning in Computer Vision**, CS6476 Course Project – Georgia Tech Aug 2024 – Dec 2024  
→ Implemented **ResNet-50** and **ResNet-101** model & tested accuracy with **PSPNet** segmentation for scene understanding  
→ Implemented **ViO** using **SIFT** features and **epipolar geometry** to estimate 6-DoF camera poses from dataset frames  
→ Implemented **Image warping** and **Panorama Stitching** using homography matrix and image coordinate transformations

**Perception-Based Intelligent Robot Localization**, Dr. Chao Shen – Carleton University May 2023 – Dec 2023  
→ Improved **JetAuto** robot's position estimates in a non-static environment through **bayesian sensor fusion** using image data  
→ Conducted **sensor fusion** using **g2o** to filter out unstable **AMCL** estimates with diverging covariances using **vision** feedback

**Vision-Based Robotic Hand Dexterity using Inverse Kinematics (IK)**, Robotics Club Mar 2023 – Jul 2023  
→ Developed an end-to-end autonomous pipeline for a robot arm to conduct object pickup using **RGB-D stereo vision** and **IK**  
→ Implemented **SE(3)** transformations using **tf2\_ros** and used **MoveIt** for planning of end-effector towards detected **6DoF** pose

**Vision-Guided Payload Pickup-Delivery using Drone**, IIT Madras Tech Meet – Flipkart GRID 4.0 Nov 2022 – Jan 2023  
→ Developed an end-to-end autonomous ROS pipeline for UAV payload pickup & drop-off scattered in a 10x10m field  
→ Implemented grid-search through **QGc** and used **OpenCV** for **6DoF** pose estimation of payloads with **ArUco** markers

## Technical Skills

**Robotics Middleware**: ROS1/ROS2, Webots, Gazebo, OpenCV, RealSense SDK, PX4, ArduPilot, MAVROS, SITL, HITL

**Algorithms & Optimization**: VLA Models, YOLO, SfM, ViO, SIFT, ORB\_SLAM2/3, Segment Anything Model, mIoUs

**Programming / Framework**: Python, C++, TF, PyTorch, MATLAB, IsaacSim, PyBullet, Eigen, SolidWorks, Unity3D, MoveIt